14.3 Evaluating Topic Models

Coherence, human checks, and what a good topic looks like.

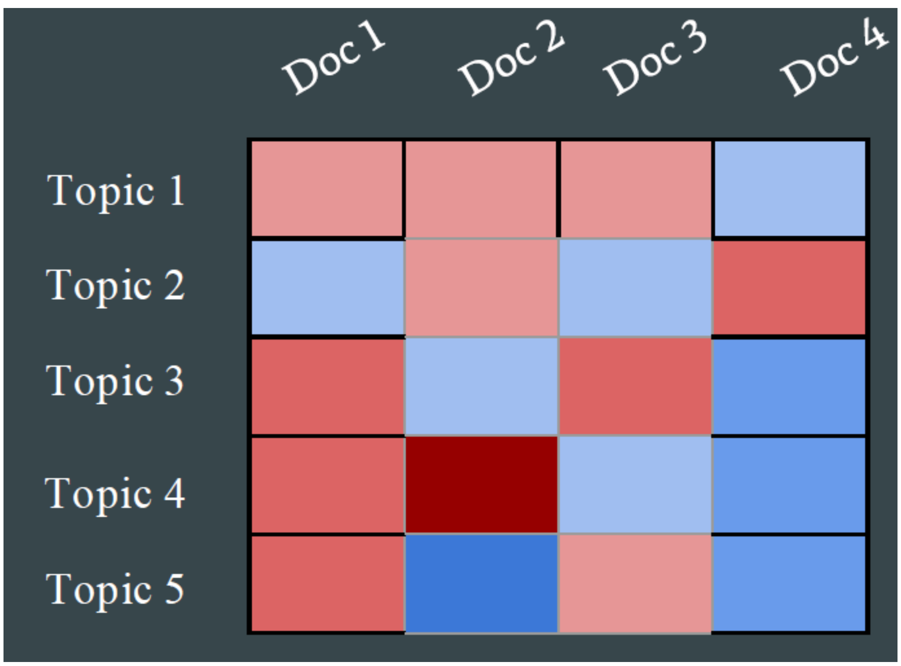
1. Why bother measuring topics

A topic model always returns (k) word lists. That does not mean the lists are themes.

Evaluation is how you:

- reject a bad run (or a bad (k))
- compare SVD, NMF, and LDA on **this** corpus
- tune priors, iterations, and preprocessing
- check that a human in the domain can name the topics
- feed the next experiment (the feedback loop)

Likelihood on the training matrix is not enough. A model can fit counts and still produce a topic that is *the, of, said* plus one noun.



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2. Automatic metrics

These do not replace reading the words. They let you discard hopeless settings before you read.

Metric	Question it asks	Watch for
Perplexity	How well does the model predict held-out documents?	Lower is better. Can disagree with human quality
Coherence (PMI, NPMI, (C_V))	Do the top words in a topic actually co-occur?	`gensim.CoherenceModel` is the usual tool
Topic diversity	Do topics share too many top words?	High diversity can also mean fragmented junk
Silhouette	If you treat (\theta) as a clustering, do documents sit in tight groups?	Needs a hard assignment; LDA is a mix
Stability	Do the same topics reappear across random seeds or subsamples?	Unstable topics are not ready to ship

Perplexity is a language-model number. Deep-learning stacks expose it; classical LDA libraries expose a bound or a held-out likelihood. Use it to compare **the same family** of models, not to declare LDA "better" than NMF.

Coherence is the metric you will actually plot against (k). If (C_V) peaks at 12 and the word lists at 12 are namable, stop there.

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3. Human and downstream checks

Topic intrusion. Show a person the top words of a topic plus one **intruder** word from another topic. If they cannot spot the intruder, the topic is a mess. There is no standard library; you can run this on a form or on Mechanical Turk.

Document clustering. If you have gold categories (section labels, product lines), treat the model's hard topic or a clustering of (\theta) as a partition. Then use ARI, NMI, or Fowlkes–Mallows (`sklearn.metrics`). This measures agreement with a known cut, not "beauty" of the words.

Use the topics. If the model is a recommender or a browsing UI, measure clicks, not NPMI. Note **recommender-systems** takes that view.

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4. How to run an evaluation, practically

1. Freeze preprocessing (stops, min count, n-grams). Then sweep (k).
2. For each (k), record coherence, diversity, and a quick read of the worst topic.
3. Rerun the winner with three seeds. If the names change, do not write a report yet.
4. Only then look at perplexity or a clustering score.

A number without a word list is not a result. A word list without a number is a vibe. You want both.

LDA, NMF, and LSA are not ranked by one universal winner. They are ranked on your documents, your (k), and whether a person can use the output.

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5. A short scorecard you can reuse

Write this down for every run:

- corpus and preprocessing hash (or a one-line description)
- method and (k), seed, iterations
- mean coherence and the worst topic's top 10 words
- a yes/no: could you name every topic in five minutes?

If the last box is no, the numbers do not matter yet. Note **14.1** and **14.2** are the models. This note is how you refuse a pretty but useless factorization. Week 15 (summarization, MT, recommenders) will reuse topics only if they pass this bar.

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6. Practice

1. Coherence is high and every topic's top words include *company*, *said*, *year*. What is broken, the metric or the preprocessing?
2. Why might the (k) that minimizes perplexity be different from the (k) a journalist can name?
3. You rerun LDA twice with (k=20). Half the topics shuffle. What do you report to a stakeholder, and what do you try next?

Report the instability. Do not average two incompatible name lists and call that the model.